AITS
ALL INDIA TEST SERIES

Dropper JEE MAIN

Additional Test

DURATION : 180 Minutes | **DATE : 16/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

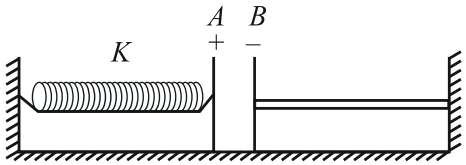
OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

Single Correct Type Questions (1-20)

1. The figure present an arrangement in horizontal plane. It consists of a parallel plate capacitor with one of its square plates fixed by means of an insulating support. While the other plate is attached to the free end of a spring made of insulating material of force constant *K*. If the capacitor carries charge *Q* and battery is disconnected, mass and side length of square plate *A* be *M* and *L* respectively, time period of oscillation of square plate *A* (assuming that it does not strike *B*) would be (ignore any friction)



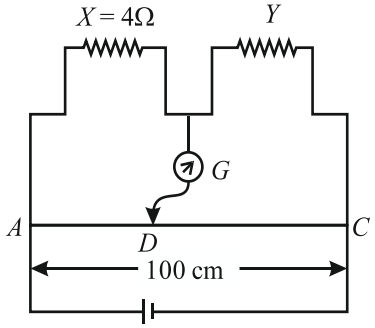
(1) $2\pi\sqrt{\frac{M}{K}}$

(2) $2\pi\sqrt{\frac{K}{M}}$

(3) $2\pi\sqrt{\frac{2M\varepsilon_0 L^3}{Q^2}}$

(4) $2\pi\sqrt{\frac{M}{K-\left(\frac{Q^2}{2\varepsilon_0 L^3}\right)}}$

2. Figure shows a metre bridge, wire *AC* has uniform cross-section. The length of wire *AC* is 100 cm. *X* is a standard resistor of 4Ω and *Y* is a coil standardised at 0°C. When *Y* is immersed in melting ice the null point is at 40 cm from point *A*. When the coil *Y* is heated to 100°C, a 78Ω resistor has to be connected in parallel with *Y* in order to keep the bridge balanced at the same point

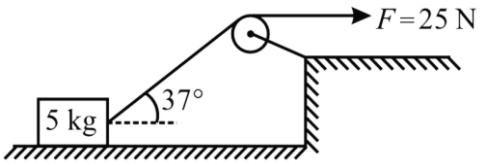


Temperature coefficient of resistance of the coil is

(1) $6.3 \times 10^{-4}\ K^{-1}$ (2) $4.3 \times 10^{-4}\ K^{-1}$

(3) $8.3 \times 10^{-4}\ K^{-1}$ (4) $2.3 \times 10^{-4}\ K^{-1}$

3. A block of mass 5kg is pulled with a constant force of F = 25 N with the help of a string. The coefficient of friction between the block and the ground is μ = 0.2. Pulley is massless and string is massless and inextensible. When the string makes an angle of 37° with the horizontal, the magnitude of acceleration of the block is



(1) 2 m/s^2

(2) 4 m/s^2

(3) 5 m/s^2

(4) 2.6 m/s^2

4. A gaseous mixture is prepared by mixing equal no. of moles of *He* and O_2 gas. Three samples of the gaseous mixture taken as *A*, *B* and *C* have initially equal volume. Now the volume of each sample is doubled. The process is adiabatic for *A*, isobaric for *B* and isothermal for *C*. If the final pressure are equal for all three samples, the ratio of their initial pressure (i.e. of *A*, *B*, *C* respectively) are

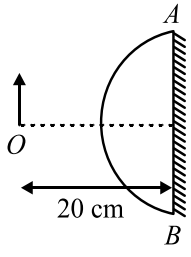
(1) $2\sqrt{2}:2:1$

(2) $2\sqrt{2}:1:2$

(3) $\sqrt{2}:1:2$

(4) $2:1:\sqrt{2}$

5. A point object is placed at a distance of 20 cm from a thin plano-convex lens of focal length 15 cm. If the plane surface is silvered, the image will form at



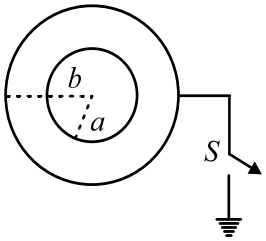
(1) 18 cm left of *AB*

(2) 12 cm left of *AB*

(3) 30 cm left of *AB*

(4) 60 cm right of *AB*

6. Two concentric conducting spheres *A* and *B* having radii *a* and *b* respectively are having a potential difference of *V* when the switch *S* is open. The potential difference when switch is closed will be



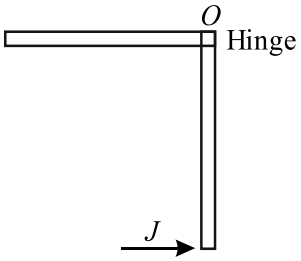
(1) Less than *V*

(2) Greater than *V*

(3) Equal to *V*

(4) Equal to $\left(\frac{V}{b-a}\right)b$

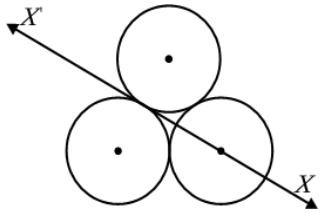
7. Two rods of mass *M* and length *L* each are rigidly joined at point *O* to form a *L*-shaped composite rod. Assembly can turn about a frictionless hinge at *O*. An impulse acts at lower end of rod. What is the angular velocity immediately after linear impulse *J* acts? Entire assembly lies in horizontal plane.



(1) $\frac{2J}{3ML}$ (2) $\frac{3J}{2ML}$

(3) $\frac{3J}{4ML}$ (4) $\frac{3J}{ML}$

8. Three discs each of mass *M* and radius *R* are placed in contact with each other as shown in figure. Then, the moment of inertia of the system about an axis *XX'* is



(1) $\frac{MR^2}{4}$

(2) $\frac{11MR^2}{4}$

(3) $\frac{2MR^2}{3}$

(4) $7MR^2$

9. A galvanometer with resistance 100Ω gives full scale deflection with a current of 10 mA. The value of shunt, in order to convert it into an ammeter of 10 ampere range, will be (nearly)

(1) 10Ω (2) 1Ω

(3) 0.1Ω (4) 0.01Ω

10. The displacement *y* (in cm) produced by a simple harmonic wave is given by

$y=\left(\frac{10}{\pi}\right)\sin\left(2000\pi t-\pi\frac{x}{17}\right)$, where *t* is time in seconds.

The maximum velocity of the particle is given by

(1) 330 m/s (2) 100 m/s

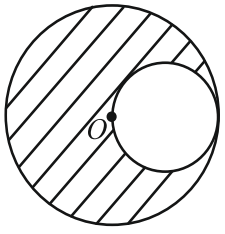
(3) 200 m/s (4) 165 m/s

11. An ideal mono-atomic gas obeys the law $T = kV^2$ where *k* is constant. The molar specific heat capacity is

(1) $\frac{5}{2}R$ (2) $\frac{7}{2}R$

(3) $2R$ (4) $\frac{5}{3}R$

12. A circular disc of radius $\frac{R}{2}$ is cut out of a bigger disc of mass *m* and radius *R* as shown in figure. Find the moment of inertia of residual disc about an axis passing through its COM and perpendicular to the plane.



(1) $\frac{121}{72}mR^2$ (2) $\frac{111}{288}mR^2$

(3) $\frac{111}{72}mR^2$ (4) $\frac{121}{288}mR^2$

13. In a region the electrostatic potential in space is given by $V=\vec{a}.\vec{r}$ where $\vec{a}$ is a constant vector and $\vec{r}$ is the radius vector of a point in the field. The electric field vector $\vec{E}$ at this point is

(1) $\vec{a}$ (2) $-\vec{a}$

(3) $\frac{\vec{a}}{2}$ (4) $-\frac{\vec{a}}{2}$

14. A chain of mass m and of length equal to one eighth of circumference of a ring of radius R, is placed upon a smooth spherical surface of radius R, with one of its ends fixed at the top of sphere. Now its upper end is released. The acceleration of the chain, immediately after releasing it, is

(1) $\frac{4g}{\pi}$ (2) $\frac{4g}{\sqrt{2}\pi}$

(3) $\frac{4g}{\pi}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$ (4) $\frac{4\sqrt{2}g}{\pi}$

15. A liquid of variable di-electric constant is filled between parallel plates of a capacitor (plate area A, plate separation d) whose value increases linearly from k to $4k$ as one moves perpendicular to plates, from one plate to the other. The capacitance of capacitor is

(1) $\frac{\varepsilon_0 kA}{3\log_e 3}$ (2) $\frac{2\varepsilon_0 kA}{3\log_e 2}$

(3) $\frac{3\varepsilon_0 kA}{2d\log_e 2}$ (4) $\frac{3\varepsilon_0 kA}{d\log_e 2}$

16. A boat moves in river water with a velocity which is two-third of river flow velocity. To minimize drifting, the boat should move, with stream direction, at an angle-equal to

(1) $90° + \cos^{-1}\left(\frac{2}{3}\right)$

(2) $90° + \cos^{-1}\left(\frac{3}{4}\right)$

(3) $90° + \sin^{-1}\left(\frac{3}{4}\right)$

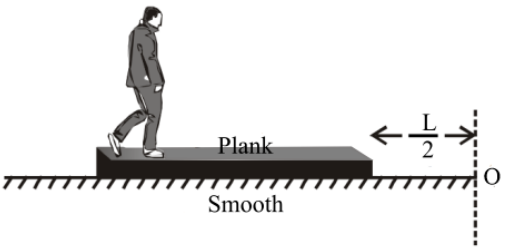
(4) $90° + \sin^{-1}\left(\frac{2}{3}\right)$

17. Three particles located at the vertices of an equilateral triangle of side l, start moving with constant speed v, heading always towards the next one so that they finally converge at the centroid. At an instant when separation between them reduces to $\frac{l}{2}$, relative normal acceleration between two particles is

(1) $\frac{9}{4}\frac{v^2}{l}$ (2) $\frac{3}{4}\frac{v^2}{l}$

(3) $\frac{3}{2}\frac{v^2}{l}$ (4) $\frac{3}{8}\frac{v^2}{l}$

18. A man of mass M walks on a plank of mass $2M$ and length L. Horizontal surface is smooth. When the man reaches the other end of the plank, he is at a distance from O equal to



(1) $\frac{L}{2}$

(2) $\frac{3L}{4}$

(3) $\frac{5L}{6}$

(4) $\frac{4L}{5}$

19. **Assertion (A):** A capacitor in series with a pure resistance bulb is put across an AC source. On introducing a dielectric inside the capacitor, the brightness of bulb reduces.

Reason (R): The impedance of capacitive circuit decreases as we increase the capacitance.

Choose the correct option.

(1) A and R both are true and R explains A.

(2) A and R both are true but R doesn't explains A.

(3) A is true but R is false.

(4) A is false but R is true

20. A wave with a frequency of 30 Hz travels along a string at a speed of 36 meters per second and reflects off a free end. How far is the first node from the free end of the string?

(1) 0.2 meters

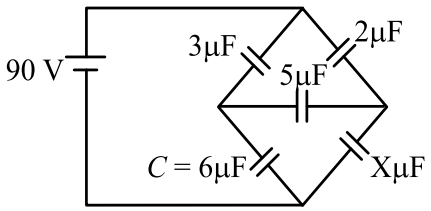
(2) 0.3 meters

(3) 0.4 meters

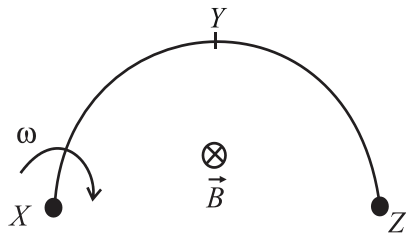
(4) 0.6 meters

Integer Type Questions (21-25)

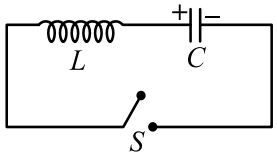
21. For what value of X (in μF) in the circuit the voltage across C = 6 μF as shown in the figure is 30 V?



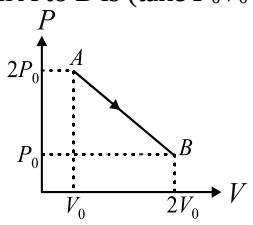
22. A metallic wire bent in the shape of semi-circle of radius R, is rotated with angular velocity ω about its end X, in a region of uniform magnetic field B into the plane of the paper as shown. The magnitude of emf induced between point Y and Z is $N \times 10^{-2}$ volt. Y is mid of the semi-circular wire XZ. Given ω = 5 radian/sec., B = 3T, R = 10 cm. Find N



23. A capacitor is charged to a potential V_0. It is connected with an inductor through a switch S. The switch is closed at time t = 0. If maximum energy stored inside the inductor is 20 J, then value of CV_0^2 is equal to _____J



24. One mole of a monoatomic gas is taken from point A to B as shown in figure. Change in internal energy from A to B is (take P_0V_0 = 100 J)



25. Consider two stationary circular loops having mutual inductance M. Starting at t = 0, the current in one of the loops is varied as $I_1 = \alpha t$ where 'α' is a constant. The current in other loop at $t = \frac{L}{R}$ is $\frac{N}{100}\frac{M\alpha}{R}$, where R is resistance of this second loop. Find 'N' (Take $\frac{e-1}{e} = 0.64$)

SECTION-II (CHEMISTRY)

Single correct Type Questions (26-45)

26. Iron(II) sulphate in acidic medium reduces permanganate to Mn^{+x}. The solution becomes yellow because of the formation of A. The yellow colour disappears if potassium fluoride is added because they form colourless complex with A.

The correct statement about the above reaction is

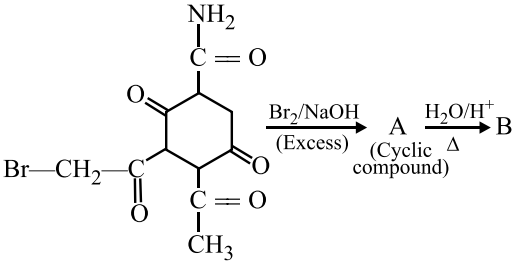
(1) In this reaction, Mn^{+7} is converted to Mn^{+4}

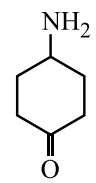
(2) A is Fe^{3+}

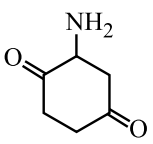
(3) A is Mn^{2+}

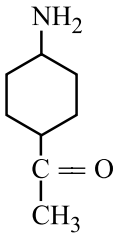
(4) In this reaction Mn^{+7} convert to Mn^{+6}

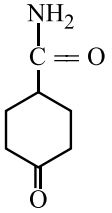
27. The major product (B) of the reaction



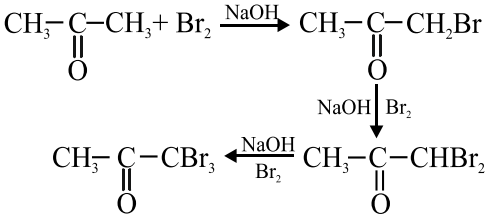
(1) 

(2) 

(3) 

(4) 

28. Bromination of acetone occurs as



The correct statement is

(1) Rate of 1st bromination is more than that of 2nd bromination

(2) Rate of 2nd bromination is more than that of 3rd bromination

(3) Rate of 3rd bromination is more than that of 1st bromination

(4) Rate of all bromination reactions is same

29. Which of the following chemical species have identical bond order?

(i) CN^{Θ} (ii) O_2^{Θ}

(iii) $NO^{\oplus}$ (iv) $CN^{\oplus}$

(1) i, ii and iii only (2) i and iv only

(3) i and iii only (4) i, iii and iv only

30. Which of the following expression represents the spectrum of Balmer series in hydrogen atom (If n is the principal quantum number of higher energy level)?

(1) $\bar{v} = \frac{R(n-1)(n+1)}{n^2}$

(2) $\bar{v} = \frac{R(n-2)(n+2)}{4n^2}$

(3) $\bar{v} = \frac{R(n-2)(n+2)}{n^2}$

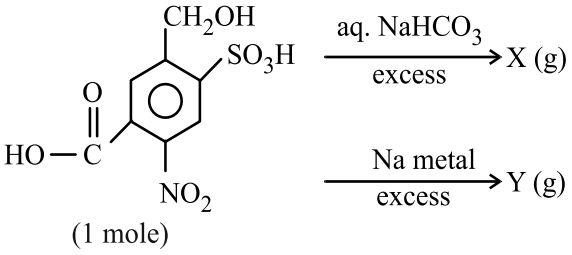
(4) $\bar{v} = \frac{R(n-1)(n+1)}{4n^2}$

31. A buffer solution is prepared in which the concentration of NH_3 is 0.30 M and the concentration of NH_4^+ is 0.20 M. pK_b for NH_3 equal to 4.74, what is the pH of this solution? (log 2 = 0.30, log 3 = 0.48)

(1) 9.08 (2) 9.44

(3) 11.72 (4) 8.73

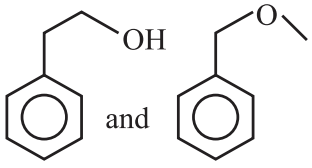
32.

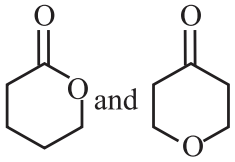


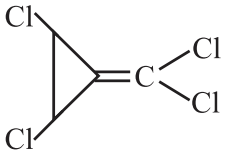
In the above reaction find out the number of moles of the gases formed in the both reactions combined.

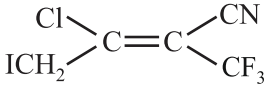
	$H_2(g)$	$SO_2(g)$	$CO_2(g)$	$NO_2(g)$
(1)	1.5	1	1	0
(2)	3	1	1	1
(3)	1.5	0	3	0
(4)	1.5	0	2	0

33. Choose the incorrect option.

(1)  and are metamers

(2)  and are functional isomers

(3)  shows geometrical isomerism

(4)  has 'E' configuration

34. When KNO_3 is heated in a closed vessel, following equilibrium is established

$$2KNO_3(s) \rightleftharpoons 2KNO_2(s) + O_2(g),\ \Delta H^\circ = +ve$$

At equilibrium

(1) Addition of KNO_3 favours the reverse (backward) reaction

(2) Addition of KNO_3 favours the forward reaction

(3) Increasing the temperature favours the forward reaction

(4) Decreasing the pressure favours the reverse (backward) reaction

35. Which one of the following arrangements of molecule is correct on the basis of their dipole moment?

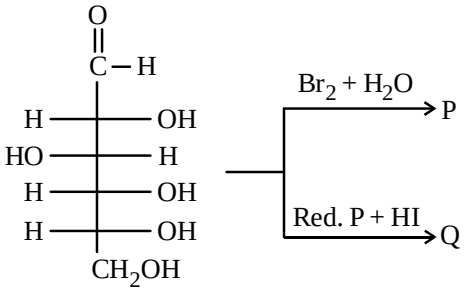
(1) $BF_3 > NH_3 > NF_3$

(2) $BF_3 > NF_3 > NH_3$

(3) $NH_3 > BF_3 > NF_3$

(4) $NH_3 > NF_3 > BF_3$

36. Consider the following reaction:



P and Q respectively are

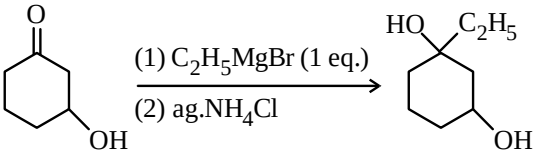
(1) Gluconic acid and n-Hexane

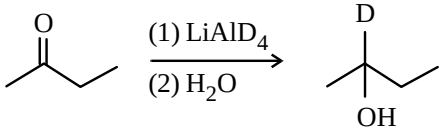
(2) Sorbitol and sorbitol

(3) n-Hexane and n-Hexane

(4) n-Hexane and sorbitol

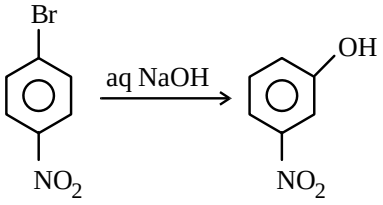
37. Which of the following reaction shows correct major product?

(1) 

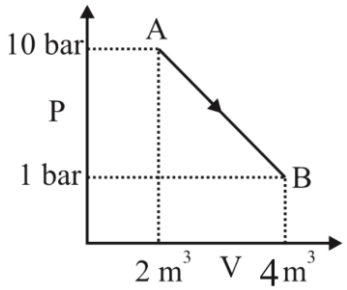
(2) 

(3) Acetaldehyde $\xrightarrow[(2)\,EtOD]{(1)\,NaBH_4}$

$CH_3-CH_2-CH_2-OD$

(4) 

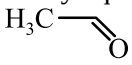
38. What is the value of ΔH_{AB} (in kJ) for an ideal gas in the given process represented in the figure? Heat supplied during the process = 2740kJ

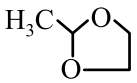


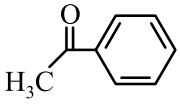
(1) 40 kJ (2) 400 kJ
(3) 1640 kJ (4) 80 kJ

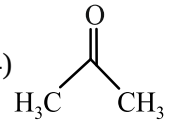
39. The shapes of PCl_4^+, PCl_4^- and $AsCl_3$ are respectively
(1) Square planar, tetrahedral, See saw
(2) Tetrahedral, See saw and pyramidal
(3) Tetrahedral, square planar and pentagonal bipyramidal
(4) Trigonal bipyramidal, tetrahedral and square pyramidal

40. Which one of the following compounds is not reduced by aq. $NaBH_4$?

(1) 

(2) 

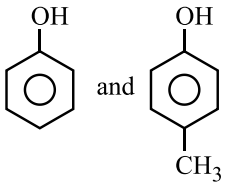
(3) 

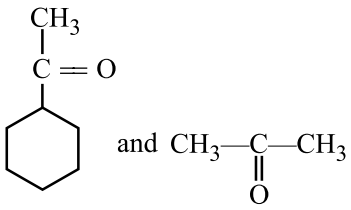
(4) 

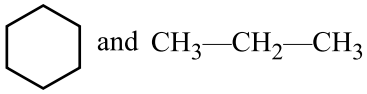
41. Cation X, Y and Z do not give precipitates with hydrochloric acid but form precipitate with hydrogen sulphide in dilute mineral acid medium. While sulphide of X, Y are insoluble in ammonium polysulphide, the sulphide of Z is soluble. Then out of the following X, Y and Z are respectively

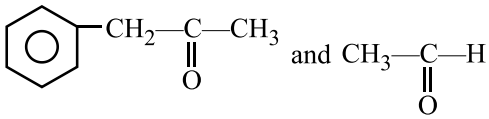
(1) $Hg_2^{2+}, Cu^{2+}, As^{3+}$
(2) Hg^{2+}, Cu^{2+}, As^{+3}
(3) Ag^+, Cu^{2+}, Hg^{2+}
(4) Ag^+, Cu^{2+}, Pb^{2+}

42. Which of the following pair of compounds can be distinguished by Tollen's test?

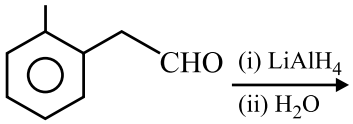
(1) 

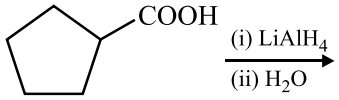
(2) 

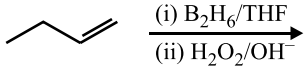
(3) 

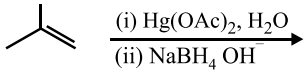
(4) 

43. Which of the following reactions will not produce 1° alcohol as product?

(1) 

(2) 

(3) 

(4) 

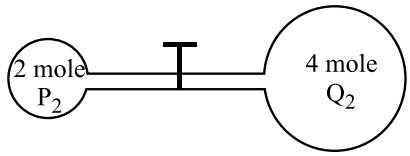
44. Identify the correct statement regarding the $[Co(NH_3)_3Cl_3]^{+3}$.
(1) It can exist in three geometrical isomers
(2) Mer-isomer is optically active
(3) Enantiomers are not possible for any isomer
(4) In facial isomer, similar groups are not present at adjacent corner of the octahedral on *l*

45. Which of the following is correct combination?

Atomic Number	Name	Symbol
(1) 105	Unnilpentium	Unp
(2) 100	Ununheptium	Uuh
(3) 109	Unnilennium	Unn
(4) 111	Unnilunium	Uuu

Integer Type Questions (46-50)

46.



The gas P_2 in the left flask is allowed to react with gas Q_2 present in right flask as

$P_2(g) + Q_2(g) \rightleftharpoons 2PQ(g)$: $K_c = 4$ at 27°C.

The concentration of PQ when equilibrium is established is C M. The value of 3C (to the nearest integer) is

47. A solution contains 2.52 g litre^{-1} of a reductant. 25 mL of this solution required 20 mL of 0.01 M $KMnO_4$ in acidic medium for oxidation. Given that each of the two atoms which undergo oxidation per molecule of reductant, suffer an increase in oxidation state by one unit. The molar mass of reductant is M then what is the value of M/2?

48. For the reaction $R - X + OH^- \longrightarrow ROH + X^-$, the rate of reaction is given as

rate = 10^{-5} [R – X][OH^-] + 10^{-6} [R – X]

If [OH^-] = 0.001 M, then the percentage of R – X reacted by S_N2 mechanism is (to the nearest integer)

49. 8 g of $Na_2CO_3.xH_2O$ are present in 500 mL of solution. This solution is diluted by water such that volume of solution becomes 1L. 100 mL of dilute solution is completely neutralised by 10 mL of 1 M HCl. The value of *x* is

50. For one molecule of B_2H_6, find the value of x + y

x = maximum number of atom that can lie in a plane.

y = total number of atoms that are sp^3 hybridised.

SECTION-III (MATHEMATICS)

Single correct Type Questions(51-70)

51. Let $f(x)$ be a differentiable function for all $x \in R$. The derivative of $f(x)$ is an even function. If the period of $f(2x)$ is 1, then the value of $f(2) + f(4) - f(6) - f(8)$ is

(1) 1 (2) 2

(3) 0 (4) Cannot be determined

52. Let

$f(x) = \sqrt{x-1} + \sqrt{x+24-10\sqrt{x-1}}; 1 \le x \le 26$ be real valued function. Then $f'(x)$ for $1 < x < 26$ is

(1) $\frac{1}{\sqrt{x-1}}$ (2) 0

(3) $2\sqrt{x-1} - 5$ (4) 1

53. Two different numbers are selected at random from 1, 2, 3, 4, ..., 100 and are multiplied, then the probability (correct up to two place of decimals) that the product thus obtained is divisible by 3, is

(1) 0.22 (2) 0.33

(3) 0.44 (4) 0.55

54. A matrix is chosen at random from the set of all matrices of order 2 with elements 0, 1 or 2. The probability that the matrix chosen is singular, is

(1) $\frac{35}{81}$ (2) $\frac{31}{81}$

(3) $\frac{32}{81}$ (4) $\frac{50}{81}$

55. The number of value(s) of x satisfying the equation

$\log_2(\log_3(\log_2 x)) \ge \sqrt{(x-8)} + \sqrt{(8-x)}$ is

(1) Zero (2) 1

(3) 3 (4) 4

56. If $f(x) = e^x$, $g(x) = x$, $x > 0$ and

$F(t) = \int_0^t f(t-x)g(x)dx$, then

(1) $F(t) = 1 - e^{-t}(1 + t)$

(2) $F(t) = e^t - (1 + t)$

(3) $F(t) = te^t$

(4) $F(t) = te^{-t}$

57. Solution to the differential equation

$$\frac{x+\frac{x^3}{3!}+\frac{x^5}{5!}+.....}{1+\frac{x^2}{2!}+\frac{x^4}{4!}+.....}=\frac{dx-dy}{dx+dy}$$ is

(1) $2ye^{2x}=C\,e^{2x}+1$

(2) $2ye^{2x}=C\,e^{2x}-1$

(3) $ye^{2x}=C\,e^{2x}+2$

(4) $y=C\,e^{2x}+2e^{2x}$

58. A series of concentric ellipses E_1, E_2,…., E_n are drawn such that E_n touches E_{n-1} at the extremities of major axis of E_{n-1} and the foci of E_n coincide with extremities of minor axis of E_{n-1}. If the eccentricity of these ellipses are equal and independent of n, then value of the eccentricity, is

(1) $\frac{\sqrt{5}}{3}$ (2) $\frac{\sqrt{5}-1}{2}$

(3) $\frac{\sqrt{5}+1}{2}$ (4) $\frac{1}{\sqrt{5}}$

59. Let g be the inverse function of f and $f'(x)=\frac{x^{10}}{(1+x^2)}$. If $g(2) = a$, then $g'(2)$ is equal to

(1) $\frac{a}{2^{10}}$ (2) $\frac{1+a^2}{a^{10}}$

(3) $\frac{a^{10}}{1+a^2}$ (4) $\frac{1+a^{10}}{a^2}$

60. Let $I_n=\int_{-\pi}^{\pi}\frac{\sin nx+2^x\cos x}{(1+4^x)\sin x}dx$, then $(n\in N)$

(1) $I_{n+2} = I_n$
(2) $I_{n+2} = 2I_n$
(3) $I_{n+2} = 3I_n$
(4) $I_{n+2} = 4I_n$

61. If $|z_1-1|\le 1, |z_2-2|\le 2, |z_3-3|\le 3$ and so on upto n terms, then the maximum value of $|z_1+z_2+\ldots z_n|$ is

(1) n
(2) $n + 1$
(3) $n(n + 1)$
(4) $\frac{n(n+1)}{2}$

62. If the variance of the data 2, 4, 5, 6, 8, 17 is 23.33, then variance of 5, 11, 14, 17, 23, 50 is

(1) 46.66 (2) 93.32
(3) 209.97 (4) 213.97

63. The point P on the parabola $y^2 = 4ax$ for which $|PR - PQ|$ is maximum, where $R(-a, 0)$, $Q(0, a)$ is

(1) $(a, 2a)$
(2) $(a, -2a)$
(3) $(4a, 4a)$
(4) $(4a, -4a)$

64. Let ABC be a triangle with vertex B(1, 2). The equation of the side BC is $y=\left(2+\sqrt{3}\right)x-\sqrt{3}$. If the incentre of the triangle ABC is (2, 3), then $\angle$ABC is

(1) 15° (2) 30°
(3) 60° (4) 75°

65. $z=x^{15}+\frac{15}{x},(x>0)$, then the minimum value of z is

(1) 14 (2) $\frac{15}{2}$
(3) 15 (4) 16

66. Last two digits of 21 to the $(100)^{th}$ power, is

(1) 00 (2) 01
(3) 11 (4) 21

67. Which of the following statement is true for the function $f(x)=\begin{cases}\sqrt{x} & x\ge 1\\ x^3 & 0\le x\le 1\\ \frac{x^3}{3}-4x & x<0\end{cases}$?

(1) $f(x)$ is monotonically increasing $\forall x\in R$
(2) $f'(x)$ fails to exist for 3 distinct real values of x
(3) $f'(x)$ changes its sign twice as x varies from $(-\infty,\infty)$
(4) Function attains its extreme values at x_1 and x_2, such that $x_1, x_2 > 0$

68. The relation R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)} on the set A = {1, 2, 3} is

(1) Reflexive but not symmetric
(2) Reflexive but not transitive
(3) Symmetric and transitive
(4) Neither symmetric nor transitive

69. Let $\vec{a},\vec{b},\vec{c}$ be vectors of length 3, 4, 5 respectively. Let $\vec{a}$ be perpendicular to $(\vec{b}+\vec{c})$, $\vec{b}$ to $(\vec{c}+\vec{a})$ and $\vec{c}$ to $(\vec{a}+\vec{b})$. Then $|\vec{a}+\vec{b}+\vec{c}|$ is

(1) $2\sqrt{5}$ (2) $2\sqrt{2}$
(3) $10\sqrt{5}$ (4) $5\sqrt{2}$

70. Let $I_1 = \int \frac{dx}{e^x + 8e^{-x} + 4e^{-3x}}$ and $I_2 = \int \frac{dx}{e^{3x} + 8e^x + 4e^{-x}}$, then value of $I_1 - 2I_2$ is equal to

(1) $\frac{1}{2}\tan^{-1}\left(\frac{e^{2x}+2}{2e^x}\right)+c$

(2) $\frac{1}{2}\tan^{-1}\left(\frac{e^{2x}+2}{e^x-1}\right)+c$

(3) $\frac{1}{2}\tan^{-1}\left(\frac{e^x}{e^x+1}\right)+c$

(4) $\frac{1}{2}\tan^{-1}\left(\frac{e^x-1}{e^x+1}\right)+c$

Integer Type Questions(71-75)

71. Six numbers are in A.P. such that their sum is 3 and the first number is four times the third number. The sum of positive terms in the A. P is

72. Let $x, y, z \in \{1, 2, 3, 4, 5\}$, then number of different values of

$$\frac{x^3}{(x-z)(x-y)} + \frac{y^3}{(y-x)(y-z)} + \frac{z^3}{(z-x)(z-y)}$$

is equal to

73. The curve $y = f(x)$ satisfies $\frac{d^2y}{dx^2} = 6x - 4$ and $f(x)$ has a local minimum value 5 when $x = 1$. Then $f(0)$ is equal to

74. Let P and Q be two points in xy-plane on the curve $y = x^7 - 2x^5 + 5x^3 + 8x + 5$ such that $\overrightarrow{OP}\cdot\hat{i} = 2$ and $\overrightarrow{OQ}\cdot\hat{i} = -2$ and the magnitude of the vector $\left|\overrightarrow{OP} + \overrightarrow{OQ}\right| = M$, (where O is origin), then the value of M is

75. The solution of the differential equation $x^2\frac{dy}{dx}\cos\frac{1}{x} - y\sin\frac{1}{x} = -1$, (when $y \to -1, x \to \infty$), is $y = \sin\frac{1}{x} - \cos\frac{1}{x} + c$ then c equals (where $c \in Z$)



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if